Vertical Integration in Uncompetitive Markets: Evidence from the Acquisition of North Star Air by the North West Company*

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Abstract

Canada's remote northern communities have some of the highest food prices and food insecurity rates in the world and rely on a small number of retail and transportation firms. In 2017, the largest retailer in the region, the North West Company (NWC), acquired a regional airline and started transporting its own air freight to some of the "fly-in" communities it serves. Using a difference-in-difference design, we evaluate the effects of vertical integration on prices and shipment quantities of perishable foods. We find that after the acquisition, communities with vertically integrated retail-air freight experienced approximately 3.9% lower prices and 9.5% higher shipment quantities relative to comparison communities. Effects are qualitatively similar in communities where the acquisition changed firm boundaries without changing routes or upstream competition (Ontario) versus communities where it created new routes and competition (Manitoba and Nunavut). Estimated price reductions are larger in communities where NWC accounts for a smaller baseline share of subsidized shipments, while total shipments increase significantly in high-share communities. A theoretical model suggests that this could be due to greater "business-stealing" and lower share of sales exposed to marginal cost reductions in more competitive communities, and evidence from retailer-level shipments supports this interpretation. Our findings show that vertical integration can benefit consumers in markets with limited competition but that the size and nature of these benefits is sensitive to upstream and downstream market structure.

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1 Introduction

Distribution plays a critical role in retail, with the journey from manufacturing plant to store accounting for a large share of value added. Large retailers have increasingly absorbed a larger share of warehousing and distribution in order to improve logistics, reduce costs (through synergies, scale or reduction of double marginalization), and gain competitive advantage against rivals. These aspects of vertical integration have been analyzed theoretically in a general setting (Spengler, 1950; Salinger, 1988; Loudermilk et al., 2023) and evaluated empirically in the context of television (Crawford et al., 2018) and gasoline (Hastings and Gilbert, 2005). In retail and distribution, Amazon represents an extreme case where the retail store is eliminated entirely, but many large retailers have regional distribution centers and their own trucks (Li, 2021). In remote communities, where distribution accounts for an even larger share of value added and there is limited competition in both the freight and the retail sector, many of these rationales for vertical integration may be amplified. This could increase the potential welfare gains from vertical integration, whether it benefits consumers depends on the extent to which marginal cost reductions are passed-through to consumers (Hong and Li, 2017) and the extent to which it promotes anti-competitive practices (e.g. actual or potential retail competitors may be excluded from distribution or receive worse pricing or service quality).

In this paper, we analyze the effects of a vertical integration episode that affected dozens of remote northern communities in Canada. The North West Company (NWC), the dominant retailer in Northern Canada and a retail monopolist in many communities, acquired North Star Air (NSA) in 2017. Before this, NWC contracted with external freight carriers (including NSA) and NSA was only providing passenger and cargo service to communities in Ontario. Subsequent to acquisition by NWC, NSA constructed a new base in Thompson, Manitoba and expanded its freight routes into Manitoba and parts of Nunavut(The North West Company, 2017, 2018; North Star Air, n.d.; Nunatsiaq News, 2018). This acquisition and subsequent developments provide a rich setting to analyze the effects of vertical integration across different market structures. In Ontario markets there was no change in routes or competition in the freight sector but the external contracts between NWC and NSA became internalized, while in Manitoba and Nunavut NSA started new routes that potentially increased competition in the air freight sector. Affected communities in Ontario and Manitoba are mostly NWC retail monopolies, but most Nunavut communities have a major retail competitor (Arctic Co-Operatives Ltd.).

Our analysis focuses on the effects of vertical integration on prices and shipments of perishable foods. Remote "fly-in" communities rely on expensive air freight for these products, making them extremely expensive given their low value to weight ratio. The federal Nutrition North Canada (NNC) program subsidizes shipments of nutritious perishable foods by air and provides data we can use to analyze the effects of vertical integration in this setting. We combine quarterly data on the cost of a food basket collected by NNC (the Revised Northern Food Basket) with monthly data on the weight of subsidized shipments, covering communities currently served by NSA and other remote communities that serve as a comparison group. Using a difference-in-differences design, our main pooled estimates indicate approximately 3.8% lower food-basket prices and 11.9% higher subsidized shipments in NSA-route communities relative to comparison communities following the acquisition and expansion. We distinguish original routes served by NSA in Ontario before the acquisition from the expansion routes into Manitoba and Nunavut. We find that price decreases and quantity

increases for both groups, with somewhat larger effects for the original routes. Empirically our findings are consistent with both reduced double-marginalization and synergies/investments that lower total marginal cost and we cannot isolate a specific mechanism. However, the effects we estimate are not immediate and take a few years after the acquisition to be realized, which suggests that the effect may be more driven by the latter.

We also examine how responses vary with NWC's baseline share of subsidized shipments, which serves as a proxy for its competitive position in local retail markets. We find larger price reductions in low-share communities, particularly on original NSA routes, while shipment increases are larger in high-share communities. We interpret this through the lens of a simple model in which NWC market dominance in a community lowers incentives to pass-through cost reductions but increases the share of retail exposed to a cost reduction, leading to ambiguous effects on quantity. In the expansion communities where NWC faces the most competition, we observe a substantial "business-stealing" effect in that shipments by competitors decrease, which supports this interpretation and indicates that retail competitors do not appear to benefit from NSA's entry into new freight routes.

We contribute to the empirical industrial organization literature on vertical integration (Hastings and Gilbert, 2005; Hortaçsu and Syverson, 2007; Lafontaine and Slade, 2007; Crawford et al., 2018) by providing the first empirical analysis (to our knowledge) of vertical integration between a retailer and airline. Our unique setting of remote communities with limited competition allows for a transparent assessment of differential impacts across varying upstream and downstream market structures. Our findings are also of intrinsic interest due to the critical importance of freight costs and food prices in Canada's remote "fly-in" communities. These communities feature the highest rates of food insecurity in the country and food prices that are often twice as high as the rest of the country in spite of Federal subsidies. The literature finds mixed evidence on whether retail market structure affects subsidy pass-through in this setting (Galloway and Li, 2023, 2026), but subsidy pass-through is regulated, unlike pass-through of other marginal cost reductions. Daley et al. (2025) find that horizontal market structure in this context does mediate the inflationary effect of cash transfers. Li et al. (2026) show that freight costs and retailer margins have contributed to higher prices in northern Canada in recent years, but it is not clear what, if anything, can help reduce these. In this context, understanding the impact of market structure and competition is critical for competition policy. Since the North West Company acquisition of North Star Air, there have been two additional mergers of freight carriers in Northern Canada reducing the total number of competing freight carriers in the Ontario, Manitoba and Nunavut fly-in communities to two. These mergers and acquisitions were approved by Canada's Competition Bureau but to our knowledge there has been no rigorous ex-post evaluation of their effects, making our findings relevant for understanding the competitive implications of future consolidation or entry in the region.

The paper proceeds as follows: we describe the institutional setting, present the conceptual framework, and discusses the data and empirical design. We then report the main results, heterogeneity and retailer-level evidence, and robustness checks. Formal derivations and additional estimates appear in the appendices.

2 Background

2.1 Retail and distribution in Northern Canada

Many communities in Northern Canada are considered "fly-in" communities, meaning that they do not have an all-weather road and rely on a combination of seasonal surface transportation and air freight for goods. Seasonal transportation includes marine transport during the open-water season and ice or winter roads during the winter months. Given the small scale of these communities, perishable food products are often flown in on small aircraft that carry a combination of passengers and cargo. The relatively low value-to-weight ratio of many food products, combined with expensive air freight, contributes to high food prices in communities facing substantial food insecurity. The Canadian government intervenes through its Nutrition North Canada (NNC) program, which subsidizes eligible products based on the weight shipped, with rates that vary by product and community (Indigenous and Northern Affairs Canada, 2017).

Most of these fly-in communities have one or two retail stores, although a handful have three or four. The dominant firm in this setting is the North West Company (NWC), a multinational, publicly listed company whose modern operations originated in the 1987 purchase of the Hudson's Bay Company's Northern Stores Division. Many of its northern stores coincide with the locations of historical trading posts. NWC also operates retail businesses in Alaska, the Caribbean, and the Pacific. At January 31, 2018, shortly after the acquisition studied here, its Canadian network included 119 Northern stores and six NorthMart stores, including stores in road-accessible communities (The North West Company, 2018). NWC has a local retail monopoly in several study communities, particularly in Northern Ontario and Manitoba. In Nunavut, it faces significant competition from Arctic Co-operatives Limited, a federation of independent, community-owned co-operatives (Arctic Co-operatives Limited, n.d.). Many communities in the Northwest Territories, Northern Quebec, and Labrador also have an NWC store, with strong competition from a separate co-operative federation in Northern Quebec. Independent stores and small chains provide additional retail alternatives in some communities.

Prior to the acquisition, North Star Air (NSA) was a regional carrier headquartered in Thunder Bay, Ontario, providing passenger and cargo service in Northern Ontario, including the contract for NWC business in the region. Its operations included bases in Pickle Lake, Red Lake, and Sioux Lookout (The North West Company, 2017; North Star Air, n.d.). Regional alternatives included Perimeter Aviation, operating from Winnipeg into Manitoba and Ontario, and Calm Air, serving Manitoba and the Kivalliq region of Nunavut. Perimeter and Calm Air belonged to the same parent company, Exchange Income Corporation (EIC). NWC stores in Manitoba and the neighbouring Kivalliq region of Nunavut were served by Calm Air on dedicated freighters prior to the acquisition. In Nunavut and the Northwest Territories, First Air and Canadian North were important passenger and freight providers before their 2019 merger (Competition Bureau, 2017, 2019).

While we can observe historic passenger routes of these airlines, scheduled passenger service is an imperfect measure of the freight alternatives available to retailers. Limited overlap between passenger networks does not necessarily imply that a retailer has access to only one freight provider: charter flights and dedicated freight contracts can provide alternatives that are not apparent from scheduled passenger routes. This distinction matters in our setting because Calm Air's NWC business in Manitoba and Kivalliq

used dedicated freight charters rather than its combination passenger-and-freight aircraft (Exchange Income Corporation, 2017), but Calm Air and Perimeter continued to operate combination aircraft (and provide a competitive freight alternative) thereafter.

One important development during this period, which may have served as a catalyst for subsequent changes in distribution, was the introduction and expansion of NNC. NNC was introduced in 2011 to replace "Food Mail," a subsidized shipping service offered through Canada Post. Under Food Mail, the government subsidized Canada Post to provide shipping rates that were generally below the commercial cost of transporting food to eligible communities. Under NNC, registered retailers and suppliers instead receive a specified subsidy per kilogram of eligible products shipped irrespective of carrier (Indigenous and Northern Affairs Canada, 2017; Crown-Indigenous Relations and Northern Affairs Canada, 2020). Eligibility expanded to additional communities, including some communities served by NSA in Manitoba and Northern Ontario, in October 2016. Further subsidy-rate increases followed in January 2019, May 2020, and February 2023 (Indigenous and Northern Affairs Canada, 2016; Galloway and Li, 2026; Nutrition North Canada, 2025). In January–March 2023, NNC-eligible communities in Ontario, Manitoba, and Nunavut together received approximately $23.8 million in subsidies, accounting for about 70.5% of the national total (Nutrition North Canada, 2025).

Because accessing the Food Mail subsidy required shipping through Canada Post, retailers sometimes made their own freight arrangements where private freight costs were lower than the subsidized postal rate, including in parts of Ontario and Manitoba. Retailers argued that receiving the subsidy directly would allow them to negotiate better freight costs, improve logistics, and lower food prices (Standing Committee, 2011). A government evaluation subsequently reported that interviewed retailers attributed improvements under NNC to their ability to negotiate freight rates directly, while noting that this opportunity remained limited in communities with few suitable carriers (Crown-Indigenous Relations and Northern Affairs Canada, 2020).

There is evidence consistent with these claims: Georgiev and Li (2026) and Crown-Indigenous Relations and Northern Affairs Canada (2020) find that the transition from Food Mail to NNC lowered prices for broad food baskets, despite being approximately budget-neutral from the federal government's perspective.[1] The transition gave retailers greater incentives to optimize their freight costs since they were no longer constrained to use Canada Post as a freight carrier, and one way of doing this may have been through acquisition of an airline.

2.2 The North West Company acquisition of North Star Air

The agreement for NWC to acquire NSA for approximately $31 million was announced on April 28, 2017. NWC also announced plans to invest a further $14 million in additional aircraft and hub capacity (The North West Company, 2017). At the time of the announcement, EIC reported that Calm Air had received 90 days' notice that NWC would transfer its freight business in Manitoba and Kivalliq to NSA. That business generated

[1] By contrast, Fafard St-Germain et al. (2019) and Daley et al. (2024) find increases in food insecurity for Food Mail eligible communities following the transition, but do not provide evidence for a retail price mechanism.

approximately $14 million in annualized revenue for Calm Air (Exchange Income Corporation, 2017).[2] The acquisition was completed on June 15, 2017, creating an integrated retailer–carrier organization (The North West Company, 2018).

The subsequent expansion involved both new destinations and investment in the freight network. A report published on August 21, 2018 described the opening of NSA's new hangar and warehouse in Thompson, Manitoba, to support service to Northern Manitoba and the Kivalliq region of Nunavut, along with several communities in the Qikiqtaaluk (Baffin) and Kitikmeot regions of Nunavut. The report also described an expansion of NSA's fleet from 11 to 16 aircraft and stated that the airline was serving approximately 50 remote communities (Nunatsiaq News, 2018). NSA's own history likewise links the acquisition to additional aircraft, the Thompson base, and expansion into Manitoba and Nunavut (North Star Air, n.d.).

Figure 1 shows the geographic distribution of the communities included in our maps and their NSA route exposure. It distinguishes 24 original NSA communities in Ontario, 26 expansion NSA communities, and 49 control communities, together with six NSA air bases. The original routes are concentrated in Northern Ontario, while the expansion routes extend into Manitoba and Nunavut (but not the far Western and Eastern parts of Nunavut). Controls are distributed across the North, including both communities near the treated regions and communities farther away. The figure identifies community locations and exposure groups; it does not draw individual flight paths or imply that every community is served directly from its nearest displayed base.

We classify as "original routes" the communities served by NSA before the acquisition. Here, the acquisition changed the ownership of an existing freight and internalized a previously external relationship between the retailer and airline without introducing new competition. The remaining treated communities are classified as "expansion routes," reflecting NSA's post-acquisition network expansion. In these communities, other freight carriers continued to operate but now faced additional competition from NSA, which offers third-party freight service as well as transporting freight for NWC The "treatment" here combines internalization of freight services with competitive entry and investment in additional capacity. Note that NSA only offers passenger service in some of these communities, and that it offers freight service to some Ontario communities that do not have an NWC store.

The expansion routes exclusively connect communities with an NWC store, as NSA only carries freight in these regions and has limited incentives to run regular freight to communities with no NWC store. For example, there is no record of regular NSA service to Whale Cove (Tikirarjuaq), which has no NWC store, despite being closer than many other NSA communities form the Thompson base. The precise timing of the expansion is less clear than the acquisition date. NWC gave notice of the Calm Air contract termination in spring 2017, while the new Thompson facilities opened in August 2018. The extent to which charters, aircraft operating from Ontario hubs, or interim Thompson operations were used between these dates is not fully documented. We use archived websites and news reports to classify service routes, and these establish that

[2]EIC described the contract as the most competitively priced/lowest margin contract in their portfolio, such that the anticipated earnings loss of the contract would be less than $1 million on $14 million in revenues. It stated that the change would not affect the efficiency or profitability of its combination-aircraft operations or its separate long-term freight contract with Arctic Co-operatives (Exchange Income Corporation, 2017).

the expansion destinations on the map were served by November 2018.[3]

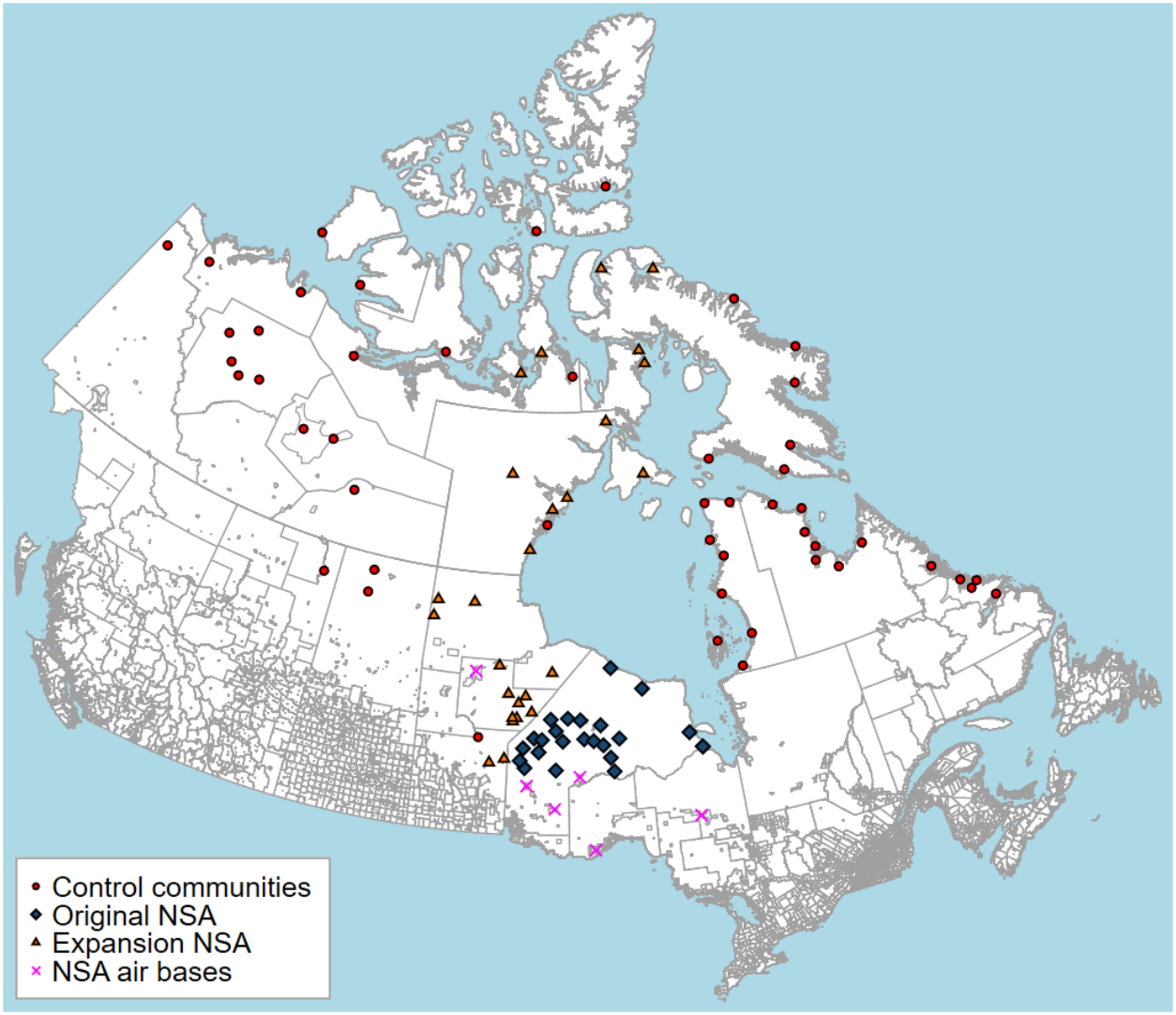


Figure 1: Study communities, NSA route exposure, and air bases

Notes: Red circles indicate 49 control communities, navy diamonds 24 original NSA communities, and orange triangles 26 expansion NSA communities. Magenta crosses identify six NSA air bases. Background polygons represent Statistics Canada's 2021 census subdivisions.

Figure 2 shows the distribution of NWC's baseline share of subsidized shipment weight across the study communities. Low-share communities have shares at or below 70%, including zero, while high-share

[3]Our archival notes record third-party cargo service to Manitoba destinations through Red Lake or Thompson by May 2018, alongside a route map listing destinations in Kivalliq and elsewhere in Nunavut. Most of the "treated" Nunavut communities in our map and classification were listed on the November 2018 third-party cargo page and in the May 2018 map, and are still listed on the website today. The exceptions are Coral Harbour and Naujaat, which first appear in a September 2024 capture. As these two Kivalliq destinations were served by the canceled Calm Air contract, are within range of the Thompson base, and have an NWC store, we classify them as treated expansion communities. On the other hand, we exclude one community that is currently listed – Kugaaruk, which did not have an NWC store until November 2023. Resolute (which has no NWC store) and Cambridge Bay (which does) were briefly listed in 2018 but were no longer listed by June 2019 and are not listed today. We classify these as untreated.

communities have shares above 70%. The high-share group includes 17 of the 24 original NSA communities, 19 of the 26 expansion communities, and 8 of the 49 controls. These shares help provide us a measure of NWC's position in the subsidized shipment market, rather than its share of total food retail sales.

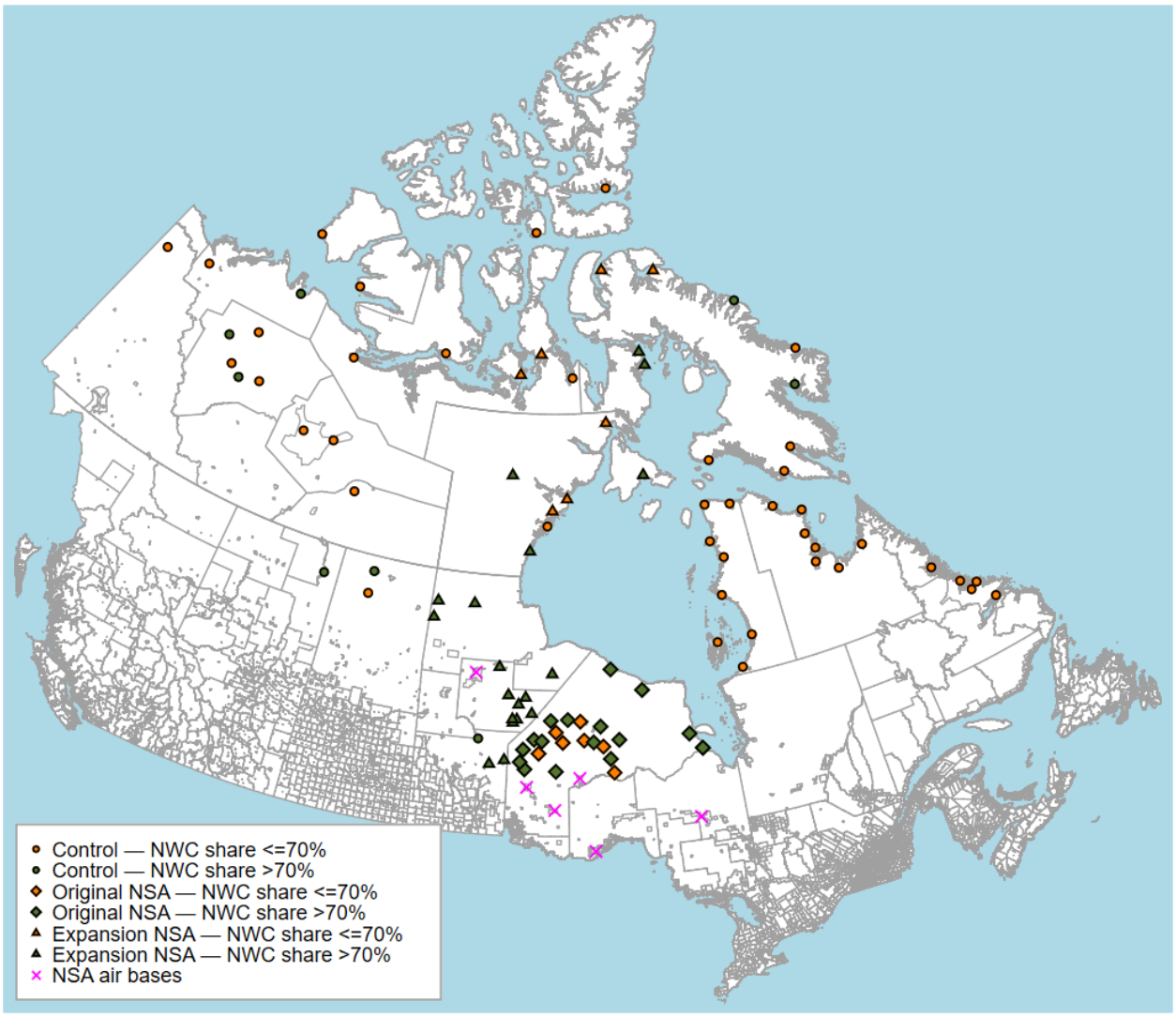


Figure 2: NSA route exposure and baseline NWC shipment share

Notes: Circles, diamonds, and triangles identify control, original NSA, and expansion NSA communities, respectively. Orange indicates baseline NWC shipment shares at or below 70%, including zero; forest green indicates shares above 70%. Magenta crosses identify NSA air bases. Background polygons represent 2021 census subdivisions.

Subsequent to the NWC–NSA acquisition, there was further consolidation among competing airlines. In 2019, Canadian North, owned by the Inuvialuit Regional Corporation, and First Air, owned by Makivvik Corporation, merged. The Competition Bureau identified substantial overlap and raised concerns that the transaction would eliminate competition in several affected markets (Competition Bureau, 2019). Following a public-interest assessment incorporating the Bureau's findings, the Government of Canada approved the

merger subject to conditions covering passenger and cargo prices, service schedules, access to northern infrastructure, and monitoring (Transport Canada, 2019). After the COVID-19 pandemic, the government and Canadian North agreed on revised conditions in April 2023. Transport Canada cited the airline's financial position and its ability to comply with the original requirements (Transport Canada, 2023).

EIC subsequently announced an agreement to acquire Canadian North for $205 million on February 24, 2025; the transaction closed on July 2, 2025 (Exchange Income Corporation, 2025). EIC, already the owner of Calm Air and Perimeter Aviation, described Canadian North's routes as complementary to its existing northern network, and the merger did not eliminate competition on any routes. However, it transferred ownership from Inuit Treaty corporations to a publicly traded corporation with potentially different incentives and substantial additional aviation assets. These transactions highlight the ongoing, potentially reactive, changes in the northern freight market, but are too recent to study here.

Taken together, our description of the institutional context and our two maps highlight that the acquisition need not have had a uniform effect across communities. Original routes primarily expose an existing network to a change in ownership, whereas expansion routes combine integration with new freight service and network investment. Within those groups, NWC's retail position also varies. These differences motivate the separate analysis of route exposure and downstream retail competition in the sections that follow.

3 Conceptual Framework

We use a retail-pricing model to examine how NWC's acquisition of North Star Air (NSA) and the airline's subsequent expansion could affect food prices and shipments. We represent the possible benefits of these changes as a reduction in NWC's cost of supplying an additional unit of food. We then compare how NWC responds when it is the only local retailer and when it faces one competitor. In each case, we compare outcomes before and after the same cost reduction, with no exit/entry. NSA enters the model through its effect on NWC's costs; we do not model the airline's decisions separately. Appendix A derives the predictions.

NWC could benefit from the acquisition in two ways. First, when the retailer and airline are separate, the freight price paid by NWC includes the airline's profit margin. Bringing them under common ownership allows NWC to consider the actual cost of carrying its goods rather than treating that margin as a payment to another company (Spengler, 1950). Although it is possible that NWC-NSA set marginal freight costs jointly to maximize total surplus through various mechanisms, if NSA set freight prices above its marginal costs this would result in a final price above the profit-maximizing price, lowering total profits. Internalization would remove this distortion. Second, better flight schedules, aircraft use, warehousing, and delivery coordination can reduce the actual cost of supplying stores. On expansion routes, new service and additional capacity may offer NWC a cheaper way to transport food. Our theoretical analysis below takes the size of the resulting cost saving as given and examines how it affects retail decisions, but in principle cost savings could be different in the original versus the expansion routes.

Each retailer sets its price to maximize profit. We assume linear demand: holding other prices fixed, equal changes in a store's price produce equal changes in how much customers buy. When two stores operate, they choose prices at the same time, and customers can switch between them. The stores can differ in location,

service, and product selection, so shoppers do not choose solely on price (Singh and Vives, 1984). Sales at each store respond more strongly to its own price than to the other store's price. The retailers can also have different costs and shares of sales. In the basic model, only NWC's cost falls. The competitor's cost and access to freight stay the same, and both stores remain open.

When NWC is the only retailer, it passes half of the cost saving to consumers through a lower price and keeps the other half as a larger profit margin on each unit sold. The cost on every unit shipped to the community decreases, and all of NWC's additional sales represent an increase in total community purchases.

When NWC faces a competitor, the two stores respond to each other's prices. NWC's lower price encourages its competitor to lower its price, and that response gives NWC a further reason to cut its own price. Under the model's assumptions, the same cost saving therefore produces a larger reduction in NWC's price when it faces competition.

However, only some of the units shipped to the community experience a reduction in costs. Some of NWC's additional sales come from customers switching stores. Although the competitor also lowers its price, NWC's price falls further, so the competitor loses sales. NWC's sales therefore increase by more than total community purchases. This can happen even though the competitor's costs and freight service have not changed, i.e. without foreclosure.

Total purchases increase in both cases, which can be interpreted as evidence for increased consumer welfare. Whether total purchases increase more under monopoly or competition depends on how customers respond to prices. The larger price reduction in competitive markets may increase NWC's sales by more, but the loss of sales at the other store may offset part of that increase. When customers' responses to price changes are the same at both stores, the model predicts a smaller increase in total purchases under competition. When those responses differ, either setting can produce the larger increase. The main point is that a larger reduction in NWC's price is not a sufficient statistic for the consumer welfare impact, and total community purchases provides a better reflection of the equilibrium effect on consumer welfare.

This distinction matters for the empirical analysis because the RNFB price observations primarily reflect NWC prices, while total shipment quantities include other retailers and purchasing channels. We therefore examine NWC's prices, its own shipments, and total community shipments separately in light of the theoretical framework described above. The basic model predicts that NWC gains sales while its competitor loses some. An extension in the appendix allows the competitor's costs to fall as well and shows how both retailers can then expand. This provides another possible explanation for changes in shipments when freight improvements benefit more than one store. The appendix also examines the broader economic effects by considering these consumer gains alongside changes in firms' profits and the costs of supplying food.

4 Data

4.1 Price and quantity data from NNC

The price outcome we analyze is the cost of the Revised Northern Food Basket (RNFB). The RNFB is a standardized, fixed-weight basket of 67 items designed to provide adequate nutrition for a family of four for one week. Nutrition North Canada (NNC) uses the basket to monitor food costs in northern communities, and

aggregates prices sent directly by retailers. RNFB costs are publicly reported at the community-by-quarter level for most NNC-eligible communities during our sample period (Nutrition North Canada, n.d.; Indigenous and Northern Affairs Canada, 2017). Approximately 75% of its total product weight consists of perishable goods that are shipped by air and eligible for the NNC subsidy, providing an informative measure of food prices most impacted by air freight costs and subsidies. However, the RNFB is not a representative price index and excludes many important foods. We use the natural logarithm or the dollar value of the RNFB cost as our main price outcomes.

The store coverage of the price data is an important consideration. Before October 2023, most independent stores and Arctic Co-operatives stores were not included in the RNFB reporting used here. Consequently, for most communities – and all treated communities – the reported prices reflect NWC rather than a comprehensive average across all local retailers.[4] The absence of comprehensive, disaggregated prices for all retailers and products limits our ability to distinguish NWC's price response from changes in community-wide prices.

The quantity outcome is constructed from NNC administrative records of subsidized shipments. These records identify shipment activity by community and reporting period and can be disaggregated by broad product category and recipient. They are more comprehensive than the RNFB price data in that they cover recorded subsidized shipments for all retailers and subsidized products, rather than prices from only from select retailers like NWC. But like the RNFB, they exclude many important food products that are not subsidy eligible.[5] We use the natural logarithm of total subsidized shipment weight or per capita kilograms of shipments (using 2016 census population counts) as the aggregate quantity outcome. In our estimation, we impose minimum shipment restrictions to exclude communities that only make minor use of the subsidy (e.g. communities that only use it during brief seasonal periods when surface transportation is not available, communities that are eligible but the retailer has not registered to receive the subsidy, and the period before October 2016 when some communities were not yet eligible).

We also use NNC administrative data to estimate the subsidy content of the RNFB or the average subsidy level, which serve as important control variables given changes in subsidy rates in October 2016, January 2019, May 2020 and February 2023. Following Galloway and Li (2023), we apply the relevant product-level subsidy rates to the corresponding weights in the basket for each community and period to generate the RNFB subsidy, and we use the total subsidy value divided by shipment weight to estimate the average subsidy per kilogram. Baseline subsidy rates also provide a proxy for relative remoteness and hub-distance in the cross-section; subsidy rates are partly determined at a regional level and are not a direct measure of cost, but are strongly correlated with observable differences in freight costs (Li et al., 2026). This allows us to flexibly control for time-varying factors that might influence communities differently depending on flight length, such as energy prices.

[4]The exceptions in our sample include Nunavik, Quebec, where the co-operative federation FCNQ is included in the community-average calculation for all 14 communities and, in three cases, represents the only store in the community. In three Northwest Territories communities, the RNFB is an average of NWC and either Stanton or Rampart. In Yukon, the sole NNC-eligible community receives special subsidies and its NWC store was replaced by a co-operative in 2014 (this does not affect our analysis). Before the 2016 fiscal year, another store in Nain, Newfoundland and Labrador, was also included in the reported average.

[5]This includes products that are subsidy-eligible when shipped by air but not eligible when shipped by surface transport, such as flour. Following the introduction of a surface-transport subsidy in January 2019, the data also record surface shipments of eligible products, although these quantities are small in the records examined for this study.

To assess retail market structure, we use the distinction between clients and recipients in the NNC shipment records. The client classification distinguishes retailers from individuals and organizations, while the recipient identifies the firm receiving the subsidy. We identify an NWC retail presence from shipments for which NWC is the recipient. Shipments from another recipient to a retail client provide evidence of competing local retail activity, rather than direct ordering by individuals or organizations.

In some cases, the recipient identifies the local store directly, as with a community co-operative. In other cases, a recipient may supply several independent stores, or different recipients may supply the same local retail market. Consequently, the number of subsidy recipients should not be interpreted mechanically as the number of local stores. We distinguish shipments associated with NWC, other local retailers, and direct orders by individuals or organizations through "Southern suppliers." This classification is important because Southern suppliers can provide an alternative to local retail purchases without necessarily operating a store in the community.

4.2 Sample composition

Table 1: Sample composition and descriptive statistics

Panel A. Sample coverage

Treatment group	Quantity-sample communities	Price-sample communities	Price observations	Quantity observations
All communities	99	89	5,219	11,727
Control communities	49	45	2,398	5,852
Original NSA	24	18	1,097	2,583
Expansion NSA	26	26	1,724	3,292

Panel B. Outcome means

Treatment group	RNFB ($ per week)	Log RNFB	Shipment kg/person	Log total shipment weight
All communities	494.98	6.19	28.27	9.86
Control communities	501.43	6.20	27.72	9.67
Original NSA	491.53	6.19	30.57	9.80
Expansion NSA	488.21	6.18	27.47	10.24

Panel C. Baseline NWC shipment-share groups

	Quantity sample			Price sample		
Treatment group	Low share	High share	High share (%)	Low share	High share	High share (%)
All communities	55	44	44.4	45	44	49.4
Control communities	41	8	16.3	37	8	17.8
Original NSA	7	17	70.8	1	17	94.4
Expansion NSA	7	19	73.1	7	19	73.1

Notes: Panels A and B report benchmark-sample counts and unweighted means. Both samples require mean and contemporaneous shipments above 15 kilograms per resident and exclude the phase-in period. Panel C groups communities by NWC's baseline share of subsidized shipment weight (October 2016–May 2017): low share is at or below 70%, including zero, and high share is above 70%. Percentages are within-sample shares of communities in the high group.

Table 1 summarizes the sample composition, outcome means, and baseline NWC shipment shares. The quantity sample covers 99 communities and 11,727 observations, while the price sample contains 89 communities and 5,219 observations. The average weekly RNFB cost is $494.98, and monthly subsidized shipments average 28.27 kilograms per resident as shown in Panel B. Panel C shows substantial differences in baseline NWC shipment shares across treatment groups. Using the 70% threshold, 70.8% of original NSA communities and 73.1% of expansion communities are classified as high-share, compared with 16.3% of controls in the quantity sample. The original-route price sample is particularly concentrated among high-share communities, with 17 of 18 communities above the threshold.

5 Empirical Strategy

5.1 Difference-in-differences

We estimate how prices and subsidized shipments change in NSA-route communities relative to control communities. The main outcomes are the natural logarithm of the RNFB cost and the natural logarithm of total subsidized shipment weight. We also examine RNFB prices in dollars and shipment quantities in kilograms per resident. The dataset is indexed by community and calendar month. Price regressions use observations for which RNFB prices are available, while shipment regressions use the monthly shipment records.

Let O_c indicate an original NSA community and E_c an expansion NSA community. These groups are mutually exclusive. The pooled NSA indicator is $A_c = O_c + E_c$; control communities have $A_c = 0$. These indicators describe community-level route exposure and are distinct from the post-treatment indicator, $Post_t$, which varies over time.

For outcome Y_{ct} in community c and period t, the pooled specification is

$$Y_{ct} = \alpha_c + \delta_t + \beta \left(A_c \times Post_t\right) + X_{ct}'\gamma + \varepsilon_{ct}, \tag{5.1}$$

where community fixed effects α_c absorb persistent differences in geography, transportation access, and retail market structure. Period fixed effects δ_t absorb seasonality and other shocks common to communities in a given calendar period. The vector X_{ct} contains the additional controls described below.

To distinguish integration on an existing network from post-acquisition expansion, we replace the pooled interaction with separate original-route and expansion-route interactions:

$$\begin{aligned} Y_{ct} = \alpha_c + \delta_t &+ \beta_O \left(O_c \times Post_t\right) \\ &+ \beta_E \left(E_c \times Post_t\right) + X_{ct}'\gamma + \varepsilon_{ct}. \end{aligned} \tag{5.2}$$

Both interactions enter the same regression, leaving control communities as the omitted route category. Thus, β_O and β_E measure the respective changes in original and expansion NSA communities relative to controls. The pooled specification summarizes the changes across both treated groups, while the disaggregated specification allows their responses to differ. The post indicator is coded to begin in August 2017, following the notice period for transferring NWC's freight business from Calm Air to NSA.

The benchmark regressions exclude the transition period from August 2017 through August 2018, when the network was being expanded and the Thompson facilities were being developed. Consequently, the retained observations compare periods up to July 2017 with periods from September 2018 onward.

The resulting coefficients describe longer-term changes associated with integration and the subsequent network configuration, rather than an immediate response on the legal acquisition date. We supplement our difference-in-difference analysis with event-study plots that include the transition period and shed light on any potential anticipatory or delayed responses of prices and quantities.

We begin with a simple specification that includes only community and period fixed effects, then add controls for changes in NNC subsidies. Price regressions control for the estimated subsidy content of the RNFB, while shipment regressions control for the community-period subsidy-per-kilogram measure which is a better indicator of the subsidy level of all subsidized shipments. Our most saturated specification also includes the interaction of each community's baseline subsidy rate (we use the level 1 NNC subsidy) with period indicators. These interactions allow the relationship between initial subsidy rates and outcomes to vary across calendar periods, and are meant to capture factors like energy costs that might vary over time and differentially affect freight costs across communities in a way that is correlated with the NSA acquisition. The benchmark sample requires shipment weight per resident to exceed 15 kilograms in the observation period and average shipment weight per resident to exceed 15 kilograms over the available community observations. These thresholds correspond to approximately the 10th percentile of the monthly shipment distribution, and we consider robustness to other thresholds.

Regressions are unweighted, and standard errors are clustered by community. The corresponding tables report observation counts and the controls included in each specification. For logarithmic outcomes, coefficients describe changes in log points and approximate proportional changes for small estimates. The level specifications express changes in dollars per RNFB basket or kilograms shipped per resident. The identifying assumption is that, conditional on the fixed effects and included controls, outcomes in NSA-route communities would have evolved like those in control communities in the absence of the acquisition and associated network changes. Persistent differences in outcome levels do not by themselves violate this assumption; differential changes unrelated to treatment would.

The interpretation also requires attention to spillovers. Changes in freight competition or NWC's wider distribution network could affect communities outside the treated route groups. If control communities are affected, the estimates measure changes relative to those affected controls rather than the full effect relative to an unaffected counterfactual. The comparison between original and expansion routes does not eliminate this concern. We therefore interpret the estimates alongside the dynamic evidence and the sensitivity of the results to alternative sample and comparison-group restrictions.

5.2 Dynamic effects/event study

We examine the timing of outcome changes using interactions between route-group indicators and calendar-period indicators. For the disaggregated specification, we estimate

$$\begin{aligned} Y_{ct} = \alpha_c + \delta_t \\ + \sum_{g \in \{O,E\}} \sum_{\tau \neq \tau_0} \beta_{g\tau} \left[G_{gc} \times \mathbf{1}\{t = \tau\} \right] \\ + X'_{ct}\gamma + \varepsilon_{ct}, \end{aligned} \tag{5.3}$$

where $G_{Oc} = O_c$, $G_{Ec} = E_c$, and τ_0 is the omitted reference period for the outcome being examined. The pooled specification replaces the separate route-group interactions with interactions using A_c. In the disaggregated specification, the original and expansion interactions enter jointly, with control communities providing the comparison group.

Th event study plots treat June 2017, when the acquisition was legalized, as the beginning of the post period. For prices, we use RNFB costs in dollars in each quarter, with March 2017 (the last price reporting period before acquisition) normalized to zero. The shipment event study uses monthly kilograms per resident and normalizes June 2017 to zero.

Each coefficient measures the change in the treated-control outcome difference relative to the relevant reference period. Unlike the benchmark before-and-after regressions, the dynamic specifications retain the transition-period observations. This allows the figures to show whether changes emerge during the network expansion or only after the Thompson facilities opened. The specifications include community and period fixed effects, the outcome-specific subsidy control, and baseline-subsidy-rate-by-period interactions. Confidence intervals are based on standard errors clustered by community.

The dynamic estimates are informative about both pre-acquisition movement and the evolution of post-acquisition differences. Note that individual months of data yield noisy estimates, which is part of the rationale for pooling before and after periods to estimate the difference-in-difference specification, and given the noise, one should not interpret the significance of individual coefficients as evidence for against pre-trends or average differences between the pre and post period (Roth et al., 2023). Systematic pre-trends would make a causal interpretation less plausible.

5.3 Heterogeneity

To examine whether responses vary with NWC's competitive position in local market market, we interact treatment with a variable S_c that captures differences across communities. For the pooled specification,

$$\begin{aligned} Y_{ct} = \alpha_c + \delta_t + \beta \left(A_c \times Post_t \right) \\ + \theta \left(S_c \times Post_t \right) + \phi \left(A_c \times Post_t \times S_c \right) \\ + X'_{ct}\gamma + \varepsilon_{ct}. \end{aligned} \tag{5.4}$$

The time-invariant terms involving A_c, S_c, and their interaction are absorbed by community fixed effects. The $S_c \times Post_t$ term allows communities with different shipment shares to experience different before-and-after changes even outside the treated group.

Our main measure of retail heterogeneity is NWC's baseline share of subsidized shipment weight, measured from October 2016 to May 2017. We define the high-share indicator S_c to equal one when this share exceeds 70% and zero when it is at or below 70%, including communities with zero NWC shipments. The treatment coefficient measures the effect in low-share communities, while the interaction coefficient measures the difference between high- and low-share communities. The high-share treatment effect is the sum of these coefficients. We also examine the robustness of these results using the continuous baseline shipment share and by excluding zero-share communities.

Finally, we examine how shipment changes are distributed between NWC and other suppliers. Using the same pooled and joint original/expansion specifications, we replace the aggregate outcome with NWC shipments per capita and competing local-retailer shipments per capita. An additional outcome includes all shipments outside NWC, encompassing both competing local retailers and direct orders. This broader non-NWC measure contains, rather than being separate from, the local-retailer component.

These decompositions distinguish NWC's own expansion from changes in other shipment channels. They also connect the empirical analysis to the conceptual framework: stronger NWC price reductions can increase NWC shipments while diverting some purchases from competitors, leaving a smaller change in aggregate shipments. A decline in rival shipments does not, by itself, identify foreclosure, as it could simply be a result of "business-stealing" due to lower prices offered by NWC. Without observations on internal pricing, freight contracts and negotiations, it is difficult to assess how NSA may have discriminated against rivals of NWC; we only know that the major regional rivals (Arctic Co-op, Fresh Market Foods) use other freight carriers.

6 Main Results

6.1 RNFB prices

Table 2 reports the price estimates. Columns (1)–(3) pool all NSA-route communities; columns (4)–(6) jointly distinguish original and expansion routes. The pooled log-price coefficient is negative throughout, falling in magnitude from −0.064 with only community and period fixed effects to −0.039 after adding the RNFB subsidy and baseline-subsidy-rate-by-period controls. The latter estimate corresponds to approximately 3.8% lower RNFB prices and is statistically significant at the 5% level. The attenuation indicates that the additional controls account for part, but not all, of the initial treated-control difference in price changes.

In the route-specific specification with all controls included, original NSA communities have an estimated price decline of 0.053 log points, approximately 5.2%, while expansion communities have a decline of 0.035 log points, approximately 3.4%. Both estimates are statistically significant. However, the larger original-route point estimate does not establish that the effects differ: the coefficient-equality test in Appendix Table B.4 has a p-value of 0.279.

Panel B provides the corresponding dollar estimates. The fully controlled pooled coefficient is −\$17.96 per weekly basket. The original- and expansion-route coefficients are −\$20.05 and −\$17.45, respectively,

significant at the 10% and 5% levels. Thus the negative price association is not specific to the logarithmic transformation. Original routes are the closer comparison to an ownership change on an existing network, whereas expansion-route estimates combine integration with entry and network investment. Neither comparison separately identifies the contribution of internalized freight margins and operating cost reductions.

Table 2: Effects of NSA route exposure on RNFB prices

	All NSA routes			Original vs. expansion		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Log RNFB price						
Post × NSA	−0.064***	−0.061***	−0.039**			
	(0.015)	(0.014)	(0.015)			
Post × original NSA				−0.094***	−0.090***	−0.053***
				(0.023)	(0.022)	(0.020)
Post × expansion				−0.051***	−0.049***	−0.035**
				(0.015)	(0.015)	(0.015)
Observations	5,219	5,219	5,219	5,219	5,219	5,219
Adjusted R^2	0.926	0.927	0.936	0.927	0.928	0.936
Panel B. RNFB price in dollars						
Post × NSA	−31.790***	−30.886***	−17.957**			
	(7.372)	(7.402)	(7.690)			
Post × original NSA				−44.083***	−42.988***	−20.052*
				(12.002)	(11.967)	(10.549)
Post × expansion				−26.293***	−25.699***	−17.446**
				(7.565)	(7.647)	(7.895)
Observations	5,219	5,219	5,219	5,219	5,219	5,219
Adjusted R^2	0.918	0.918	0.930	0.919	0.919	0.930
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
RNFB subsidy control	No	Yes	Yes	No	Yes	Yes
Baseline rate × period	No	No	Yes	No	No	Yes

Notes: Columns (1)–(3) pool all NSA route communities. Columns (4)–(6) include the original- and expansion-route interactions jointly. Our sample restricts to periods with above 15 kilograms of shipments per capita and communities with an average of at 15 kilograms per capita, and excludes the phase-in period. All specifications include community and period fixed effects. Columns (2), (3), (5), and (6) control for the estimated RNFB subsidy; columns (3) and (6) additionally include baseline-rate-by-period interactions. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

6.2 Subsidized shipment quantities

Table 3 reports the aggregate shipment estimates. In Panel A, the pooled log-weight coefficient declines from 0.107 in column (1) to 0.095 in column (3), corresponding to approximately 9.5% higher shipments in the specification with all controls included. The original-route coefficient is 0.124 (approximately 12.4%), while the expansion-route coefficient is 0.083 (approximately 8.3%) in column (6). All three fully controlled estimates are statistically significant at the 1% level.

Table 3: Effects of NSA route exposure on subsidized shipment quantities

	All NSA routes			Original vs. expansion		
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Log total shipment weight						
Post × NSA	0.107***	0.097***	0.095***			
	(0.021)	(0.020)	(0.022)			
Post × original NSA				0.141***	0.126***	0.124***
				(0.032)	(0.031)	(0.033)
Post × expansion				0.087***	0.082***	0.083***
				(0.023)	(0.022)	(0.023)
Observations	11,727	11,727	11,727	11,727	11,727	11,727
Adjusted R^2	0.963	0.963	0.965	0.963	0.963	0.965
Panel B. Shipment kilograms divided by 2016 census population						
Post × NSA	3.077***	2.848***	2.859***			
	(0.658)	(0.654)	(0.665)			
Post × original NSA				4.393***	4.088***	4.225***
				(0.874)	(0.893)	(0.906)
Post × expansion				2.284***	2.196***	2.297***
				(0.739)	(0.723)	(0.717)
Observations	11,727	11,727	11,727	11,727	11,727	11,727
Adjusted R^2	0.619	0.620	0.639	0.621	0.621	0.640
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Subsidy/kg control	No	Yes	Yes	No	Yes	Yes
Baseline rate × period	No	No	Yes	No	No	Yes

Notes: Columns (1)–(3) pool all NSA routes; columns (4)–(6) include original and expansion routes jointly. Our sample restricts to periods with above 15 kilograms of shipments per capita and communities with an average of at 15 kilograms per capita, and excludes the phase-in period. All specifications include community and period fixed effects. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Panel B shows positive changes in shipment kilograms per resident for both route groups. The estimates with all controls included are are 2.859 kilograms per resident for pooled NSA exposure, 4.225 for original routes, and 2.297 for expansion routes, all significant at the 1% level. The logarithmic and per-capita specifications provide complementary measures: one measures proportional changes in total shipment weight, while the other measures absolute changes per resident.

All specifications use the same 11,727 observations from 99 communities. Although the estimated proportional increase is larger on original routes, the coefficient-equality test in Appendix Table B.4 does not reject equality between the original- and expansion-route effects ($p = 0.196$).

6.3 Dynamic evidence

Figures 3 and 4 show the evolution of adjusted outcome differences for all NSA cargo routes, original NSA routes, and expansion routes. The panels labeled "New NSA routes" correspond to the expansion communities in Manitoba and Nunavut. The price figure reports quarterly changes in RNFB costs in dollars, while the quantity figure reports changes in monthly shipment kilograms per resident.

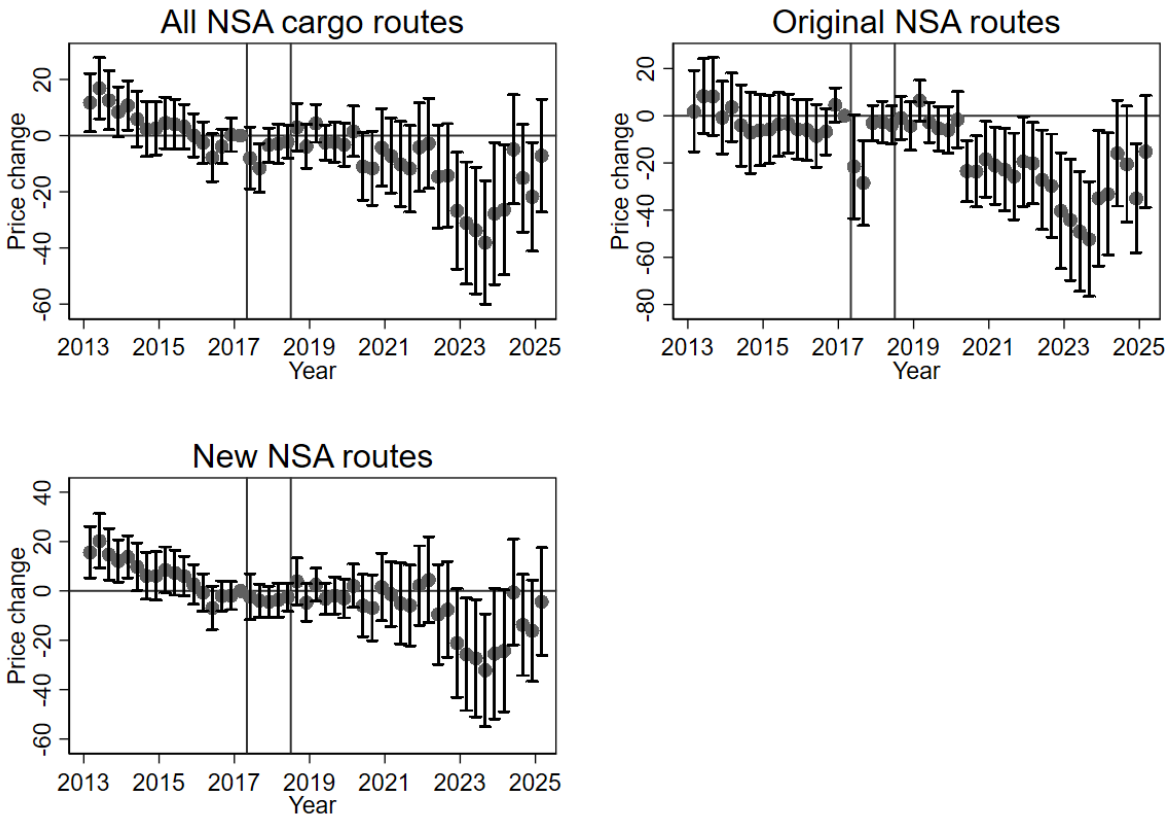


Figure 3: Dynamic differences in RNFB prices

Notes: Event-study coefficients measure quarterly changes in RNFB prices (dollars) relative to March 2017. Panels distinguish pooled, original, and expansion NSA routes. Regressions include community and period fixed effects, RNFB subsidy controls, and baseline-rate-by-period interactions. The phase-in period is retained. The vertical lines mark NWC's acquisition of North Star Air in June 2017 and the opening of the Thompson air base in August 2018. Error bars represent 95% confidence intervals based on community-clustered standard errors.

The price estimates in Figure 3 do not show a single, permanent downward shift at the time of legal acquisition. The pooled and expansion-route coefficients fluctuate around zero through much of the acquisition and initial expansion interval, with more pronounced negative estimates appearing later. Original-route estimates show an initial decline and partial reversal before a more sustained reduction later in the sample. The deepest subsequent price reductions appear in the original-route panel. The expansion-route estimates also become negative, but their confidence intervals are often wider relative to the estimated changes. All three series recover partially from their late-sample troughs, indicating that the price response is not monotonically increasing in magnitude. Formal pre-trend tests reported in Appendix Table B.8 compare average coefficients across consecutive pre-treatment windows of 3, 6, and 12 months. None of the tests rejects equality at the 5% level, although three price tests reject at the 10% level.

Figure 4 shows a more distinct change in shipment quantities after the acquisition and network expansion interval. Before that interval, the coefficients fluctuate within a comparatively narrow range without a

visually pronounced, sustained upward progression. Larger positive estimates emerge after the second vertical marker (indicating all NSA markets are being served) in all three panels. The increases reach their highest values during the middle of the post-acquisition period, then moderate while generally remaining above the reference-period difference.

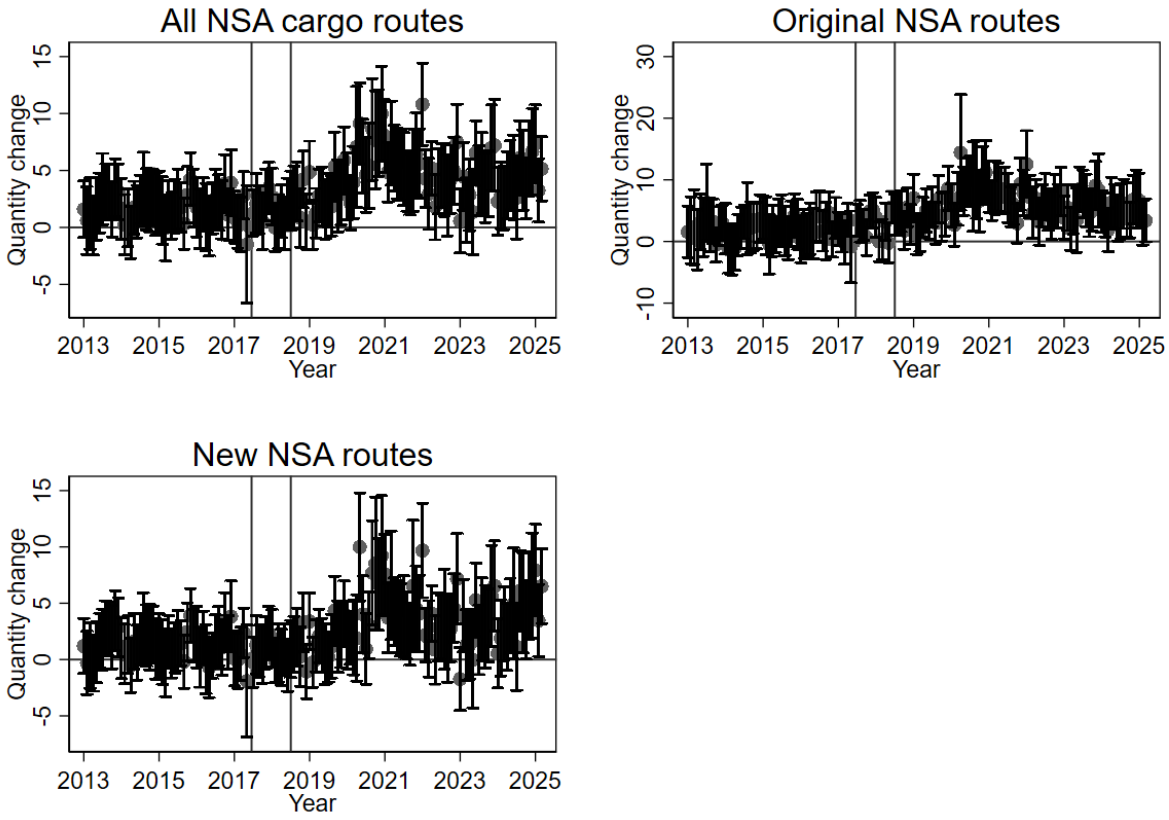


Figure 4: Dynamic differences in shipment quantities

Notes: Event-study coefficients measure monthly changes in subsidized shipment kilograms per resident relative to June 2017. Panels distinguish pooled, original, and expansion NSA routes. Regressions include community and period fixed effects, per-kilogram subsidy controls, and baseline-rate-by-period interactions. The phase-in period is retained. The vertical lines mark NWC's acquisition of North Star Air in June 2017 and the opening of the Thompson air base in August 2018. Error bars represent 95% confidence intervals based on community-clustered standard errors.

Both original and expansion routes contribute to this pattern. Original-route point estimates are larger at some dates, while expansion-route estimates fluctuate substantially during the later period. The quantity evidence is better described as a delayed increase followed by continued variation than as an immediate, uniform jump at the acquisition.

Taken together, the figures are consistent with lower relative RNFB prices and higher relative shipments developing over time rather than appearing simultaneously at the transaction date. This timing is compatible with a gradual adjustment in freight arrangements and network capacity. It does not separately identify the

effects of ownership, entry, or operating efficiencies, but does appear to rule out an immediate effect operating through elimination of large double-marginalization effects in the original communities.

7 Heterogeneity and Retailer-Level Responses

7.1 Baseline NWC shipment share

Table 4 examines whether price and shipment responses differ between communities with high and low baseline NWC shipment shares. High-share communities are those where NWC accounts for more than 70% of baseline subsidized shipment weight; low-share communities have shares at or below 70%, including zero. Columns (1)–(2) report the treatment effects for low-share communities and the additional effects associated with a high share. Columns (3)–(4) report the corresponding treatment effects for high-share communities, calculated by adding each treatment coefficient to its high-share interaction. Column (3) uses the pooled regression in column (1), while column (4) uses the original/expansion regression in column (2).

Panel A shows significant price reductions in low-share communities. The pooled coefficient is -0.064, while the original- and expansion-route coefficients are -0.107 and -0.062, respectively. The original-route estimate is significant at the 1% level, and the pooled and expansion-route estimates are significant at the 5% level. The corresponding high-share effects are smaller in magnitude, at -0.013, -0.030, and -0.008, and none is statistically distinguishable from zero at conventional levels. Thus we only observe quantitatively large and statistically significant price reductions in the low share communities.

Panel B reports positive shipment effects in both high and low NWC market share communities. Among low-share communities, the pooled coefficient is 0.048, the original-route coefficient is 0.116, and the expansion-route coefficient is 0.019. Only the original-route estimate is statistically significant at the 10% level. Among high-share communities, the corresponding joint effects are 0.118, 0.138, and 0.114, all significant at the 5% level.

On original NSA routes, price reductions are significantly smaller in high-share communities. The original route price interaction remains positive and statistically significant. Shipment increases are statistically significant in high-share communities across the pooled and route-specific specifications. Although this could theoretically result from different market-level price elasticities in each community, recall that we only observe NWC prices in treated communities but our quantity measure includes all NNC subsidized air freight to the community. In low share communities, the reduction in NWC prices affects a much smaller share of initial sales, and if competitors do not reduce their prices by a similar magnitude, there is no guarantee that the overall community price (weighted by store expenditure shares) fell by more in these communities. The NWC price decline could be lower in high share communities but this decline affects every unit sold in the community, which could lead to a larger quantity response. Moreover, in low share communities there is potential for business-stealing whereby NWC price reductions lower shipment quantity for competitors, which offsets some of the additional sales generated by NWC when it lowers prices in response to the NSA acquisition. The original-route findings are broadly consistent when using a continuous baseline NWC shipment share or excluding zero-share communities (i.e. those without an NWC store). The price-share interaction remains positive and statistically significant, while estimated shipment effects remain positive, as

shown in Appendix Tables B.9 and B.10.

Table 4: Heterogeneity by baseline NWC shipment share

	Regression coefficients		High-share joint effects	
	All NSA	Original/expansion	All NSA	Original/expansion
	(1)	(2)	(3)	(4)
Panel A. Log RNFB price				
Post × NSA	−0.064**		−0.013	
	(0.026)		(0.021)	
Post × NSA × high share	0.051			
	(0.033)			
Post × original NSA		−0.107***		−0.030
		(0.017)		(0.024)
Post × original × high share		0.077***		
		(0.024)		
Post × expansion		−0.062**		−0.008
		(0.026)		(0.021)
Post × expansion × high share		0.054		
		(0.035)		
Observations	5,219	5,219		
Adjusted R^2	0.937	0.937		
Panel B. Log total shipment weight				
Post × NSA	0.048		0.118**	
	(0.031)		(0.055)	
Post × NSA × high share	0.071			
	(0.063)			
Post × original NSA		0.116*		0.138**
		(0.061)		(0.062)
Post × original × high share		0.021		
		(0.082)		
Post × expansion		0.019		0.114**
		(0.031)		(0.056)
Post × expansion × high share		0.095		
		(0.065)		
Observations	11,727	11,727		
Adjusted R^2	0.965	0.965		
Community fixed effects	Yes	Yes		
Period fixed effects	Yes	Yes		
Outcome-specific subsidy	Yes	Yes		
Baseline rate × period	Yes	Yes		

Notes: Baseline NWC share is measured using subsidized shipment weight from October 2016 to May 2017. High share exceeds 70%; low share is at or below 70%, including zero. Missing shares are excluded. Columns (1)–(2) report low-share treatment coefficients and high-minus-low interactions; columns (3)–(4) report high-share joint effects. Regressions include community and period fixed effects, outcome-specific subsidies, and baseline-rate-by-period controls. The 15-kilogram restrictions apply and the phase-in period is excluded. Standard errors are clustered by community; $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

7.2 Retailer-level shipment decomposition

Table 5 decomposes shipment changes into NWC and non-NWC components, using kilograms per resident. The pooled estimate is an increase of 2.859 kilograms in total shipments and 3.151 kilograms in NWC shipments, both significant at the 1% level. The estimates for competing local retailers and the broader non-NWC channel (local retailers plus direct orders by individuals or establishments) are negative but not significantly different than zero.

Table 5: Retailer-level decomposition of shipment quantity effects

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
Panel A. All NSA routes				
Post × NSA	2.859***	3.151***	−0.464	−0.292
	(0.665)	(0.733)	(0.653)	(0.685)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.639	0.912	0.865	0.842
Panel B. Original vs. expansion routes				
Post × original NSA	4.225***	2.301**	1.669*	1.924**
	(0.906)	(1.110)	(0.858)	(0.939)
Post × expansion	2.297***	3.500***	−1.342**	−1.203*
	(0.717)	(0.733)	(0.612)	(0.633)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.640	0.912	0.867	0.843
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: Outcomes are monthly shipment kilograms per resident. Column (4) includes the local-retailer shipments in column (3) and direct orders by individuals and establishments. Panel A pools NSA routes; Panel B distinguishes original and expansion routes jointly. Regressions include community and period fixed effects, per-kilogram subsidies, and baseline-rate-by-period controls, with the 15-kilogram restrictions and no phase-in observations. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The route-specific results show that pooled estimates conceal substantial differences in the composition of shipment changes. On original routes, NWC shipments increase by 2.301 kilograms per resident, while competing local-retailer shipments increase by 1.669.[6] The broader non-NWC increase is 1.924 kilograms. The NWC and broader non-NWC coefficients are significant at the 5% level, and the local-retailer coefficient at the 10% level. The estimated total increase of 4.225 kilograms therefore reflects expansion in both NWC and non-NWC shipments, rather than displacement of competitors by NWC alone.

On expansion routes, NWC shipments increase by 3.500 kilograms per resident, while competing local-retailer shipments decline by 1.342 and broader non-NWC shipments decline by 1.203. These coefficients

[6]Note that most local-retailers in this region serve communities without a Northwest company store. The largest direct order competitor – Fresh Market Foods – uses Bearskin Airline, not North Star Air.

are significant at the 1%, 5%, and 10% levels, respectively. The resulting total increase is 2.297 kilograms: non-NWC contraction offsets part of NWC's expansion. This pattern is consistent with business reallocation alongside market expansion, but it does not identify foreclosure or establish why rival shipments fall.

Appendix Table B.1 assess whether the displacement effect we observe is driven by the observed change in RNFB price for NWC, by including price as a control. Including the price variable as a control does decrease the extent of displacement on expansion routes and overall, but Table B.2 shows that this may be largely an artifact of using the smaller estimation sample (dropping some communities and time periods) when including prices rather than inclusion of the price variable itself. We report log quantity specifications in Appendix Table B.3, which are positive for both NWC and other-retailer shipments; the log effects are not additive and magnitudes cannot be compared to each other directly since they apply to communities that vary significantly in size and retail market structure.

8 Robustness

8.1 Time windows and comparison groups

Table 6 varies the sample while using the same baseline specification with all controls. Column (2) starts in June 2015 rather than the beginning of the available panel. Column (3) retains the phase-in period, and column (4) excludes observations from 2022 onward. Columns (5)–(6) retain the NSA-treated communities but narrow the controls to Quebec and Nunavut, or Quebec alone.

The price estimates remain negative throughout. The pooled coefficient changes from -0.039 in the benchmark to -0.027 when early observations are removed, -0.038 when the phase-in period is included, and -0.023 when observations from 2022 onward are dropped. These estimates remain statistically significant, although the shorter pre-period estimate is significant only at the 10% level. Restricting the comparison group produces pooled estimates of -0.042 for Quebec and Nunavut and -0.084 for Quebec alone. Original-route price coefficients are negative and significant in all six columns. Expansion-route estimates are also negative throughout, but the June-2015-onward estimate is not statistically significant. The sign of the price association is stable across these checks, while its magnitude remains sensitive to the comparison group.

The quantity results in this table use the separate 11,727-observation benchmark identified above. Relative to its pooled coefficient of 0.095, the time-window variants produce coefficients of 0.104, 0.080, and 0.095, all significant at the 1% level. Narrowing the controls to Quebec and Nunavut reduces the pooled estimate to 0.052, still significant, while using Quebec alone reduces it to 0.022, which is not significant. Under Quebec-only controls, neither the original-route nor the expansion-route quantity estimate is distinguishable from zero. The evidence for shipment growth is therefore more dependent on the comparison group than the evidence for lower RNFB prices.

Table 6: Robustness to sample restrictions and comparison groups

	Benchmark	June 2015 onward	Include phase-in	Before 2022	Quebec + Nunavut	Quebec only
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A. Log RNFB price: all NSA routes						
Post × NSA	−0.039**	−0.027*	−0.038***	−0.023**	−0.042**	−0.084***
	(0.015)	(0.014)	(0.014)	(0.009)	(0.017)	(0.017)
Observations	5,219	4,626	5,523	2,281	4,379	3,689
Adjusted R^2	0.936	0.935	0.936	0.831	0.941	0.945
Panel B. Log RNFB price: original vs. expansion routes						
Post × original NSA	−0.053***	−0.045***	−0.050***	−0.030**	−0.056**	−0.100***
	(0.020)	(0.017)	(0.018)	(0.013)	(0.022)	(0.023)
Post × expansion	−0.035**	−0.022	−0.034**	−0.020**	−0.039**	−0.081***
	(0.015)	(0.015)	(0.014)	(0.009)	(0.018)	(0.018)
Observations	5,219	4,626	5,523	2,281	4,379	3,689
Adjusted R^2	0.936	0.935	0.936	0.832	0.942	0.946
Panel C. Log shipment quantity: all NSA routes						
Post × NSA	0.095***	0.104***	0.080***	0.095***	0.052***	0.022
	(0.022)	(0.021)	(0.020)	(0.021)	(0.019)	(0.022)
Observations	11,727	9,601	12,916	8,018	9,451	7,748
Adjusted R^2	0.965	0.964	0.965	0.968	0.962	0.960
Panel D. Log shipment quantity: original vs. expansion routes						
Post × original NSA	0.124***	0.137***	0.107***	0.134***	0.071**	0.026
	(0.033)	(0.029)	(0.030)	(0.030)	(0.029)	(0.030)
Post × expansion	0.083***	0.089***	0.069***	0.078***	0.047**	0.021
	(0.023)	(0.022)	(0.021)	(0.022)	(0.020)	(0.023)
Observations	11,727	9,601	12,916	8,018	9,451	7,748
Adjusted R^2	0.965	0.964	0.965	0.968	0.962	0.960
Community fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Relative to the benchmark in column (1), column (2) begins in June 2015, column (3) includes the phase-in period, and column (4) excludes 2022 onward. Columns (5)–(6) restrict controls to Quebec and Nunavut or Quebec alone, retaining all treated communities. All specifications include community and period fixed effects, outcome-specific subsidies, and baseline-rate-by-period controls. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

8.2 Shipment cutoffs, outcome coverage, and seasonality

Appendix Table B.5 varies the restrictions on both average and contemporaneous shipment kilograms per resident. Pooled price estimates are similar across no cutoff and cutoffs of 10, 15, and 20 kilograms, ranging from −0.035 to −0.039. Quantity magnitudes are more sensitive: the pooled log coefficient declines from 0.125 with no cutoff to 0.066 at 20 kilograms, although it remains significant at the 1% level in every column. The original-route quantity estimate decreases from 0.223 to 0.093 across the same extremes. These comparisons indicate some sensitivity of the effect on shipments to including or excluding communities with below average shipment volumes.

Appendix Table B.6 replaces the benchmark shipment outcome with the broader measure covering all NNC shipments and uses its corresponding subsidy control. On the same reported 11,727 observations, the pooled coefficient increases from 0.095 to 0.130. The original- and expansion-route estimates also increase, from 0.124 to 0.183 and from 0.083 to 0.107. All are significant at the 1% level. Thus the positive estimates in this sample are not confined to the benchmark shipment definition.

Appendix Table B.7 allows communities to have their own recurring quarter-of-year patterns. The pooled price coefficient changes from -0.039 to -0.038, and the pooled log-quantity coefficient from 0.095 to 0.090. Original- and expansion-route coefficients also change little.

Overall, the price results are consistently negative across the reported sample and specification checks. Shipment estimates are generally positive, but their magnitude and precision depend more on the shipment cutoff and control group.

9 Conclusion

This paper shows how the North West Company's acquisition of North Star Air and the airline's subsequent expansion changed food prices and shipments in remote Canadian communities. Although none of the markets we analyze can be described as highly competitive, the acquisition generated distinct changes in competition. In the upstream freight sector, the original Ontario routes experienced a change in ownership but no entry, while the expansion Manitoba/Nunavut markets experienced a new entrant, albeit one that has different competitive incentives than existing firms. In the downstream retail sector, NWC faced varying degrees of competition in affected markets. Our analysis suggests that on average all communities with vertically integrated air freight and retail benefited through lower prices, resulting in increased shipments of subsidized food and implying significant consumer welfare gains. However, the mechanisms and magnitudes varied with market structure, with some evidence of larger benefits in communities where NWC faced less retail competition. Our theoretical model and retailer-level evidence suggest that the acquisition lead to larger cost reductions for NWC, leading to some business stealing and a smaller cost reductions (at the community level) in the more competitive expansion markets. Although we cannot nail down a precise mechanism, the delayed effects of the acquisition observed in our event study graphs suggest that immediate reductions in upstream profit margins are unlikely to be the main factor driving our results. Investments in capacity and improvements in freight operations seem more likely but confirming this would require further investigation.

Our findings suggest that vertical integration can benefit consumers even in markets with limited competition and even when it results in some rival displacement. However, our analysis is limited by the lack of data on the full range of products sold, the terms of freight contracts, and prices charged by competing retailers. These gaps limit our ability to measure the underlying cost savings and determine how their benefits are distributed between consumers, NWC, and other firms. Moreover, the earlier transition from Food Mail to NNC that preceded the NSA acquisition, and the subsequent airline mergers, show that firms' responses to policy and ownership changes can unfold over many years. Strategies intended to build a lasting competitive advantage may not be fully reflected in the period we observe, which should not be viewed as a long-term equilibrium. Future work should examine the effects of subsequent mergers and whether the

observed price and shipment changes persist as firms adjust. It should also consider other forms of vertical integration in northern freight, including marine transportation, where NWC and the co-operatives have shipping interests, and trucking to communities served by seasonal winter roads. Data on product quality, freshness, and availability would provide additional measures of the benefits reaching consumers. Passenger fares and service on NSA's Ontario routes also warrant attention, since changes in ownership and investment can affect both freight and passenger operations. Together, these questions would extend the analysis beyond food prices and shipment volumes to the broader, longer-term effects on northern communities.

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Appendix

A Retail Competition, Cost Pass-Through, and Shipments

A.1 Setup

Air freight is the upstream input; food retailing is the downstream activity. Retailer 1 is NWC and retailer 2 is a local competitor. Each retailer chooses its price p_i to maximize operating profit $\pi_i = (p_i - c_i)q_i$, where q_i is sales and c_i is the constant cost of supplying one additional unit, including freight. Quantities use a common unit of food, so community purchases are $Q = q_1 + q_2$. Airline pricing and freight profits earned on the rival's shipments are outside the model.

The acquisition and subsequent expansion of NSA enter through a reduction in NWC's delivered cost. Superscripts 0 and 1 denote before and after this reduction:

$$c_1^1 = c_1^0 - \delta, \qquad c_2^1 = c_2^0, \qquad \delta > 0. \tag{A.1}$$

The saving can reflect an internalized freight margin or lower resource costs from logistics and capacity improvements. Its size is taken as given and can differ between original and expansion routes. The rival's freight access and cost remain unchanged. We compare the same δ under monopoly (M) and duopoly (D); within each case, demand parameters and the number of stores are fixed. Write $\Delta x = x^1 - x^0$. Active stores have positive prices and sales throughout, with no entry or exit.

We use the linear differentiated-demand model of Singh and Vives (1984). Community demand follows from maximizing gross benefit less expenditure, $V - p_1q_1 - p_2q_2$, where

$$V(q_1, q_2) = a_1q_1 + a_2q_2 - \frac{1}{2}\left(b_1q_1^2 + 2dq_1q_2 + b_2q_2^2\right). \tag{A.2}$$

Here $a_i > 0$ measures willingness to pay, $b_i > 0$ governs its decline with own purchases, and $0 < d < \min\{b_1, b_2\}$ captures substitution between stores. Money has constant marginal utility. The purchase conditions $\partial V/\partial q_i = p_i$ give inverse demands

$$p_1 = a_1 - b_1q_1 - dq_2, \qquad p_2 = a_2 - b_2q_2 - dq_1. \tag{A.3}$$

Different intercepts, slopes, and costs allow unequal store sales. When only NWC is available, setting $q_2 = 0$ recovers its monopoly demand from the same preferences.

A.2 Monopoly

NWC faces $q_1 = (a_1 - p_1)/b_1$ and solves

$$\max_{p_1} \; (p_1 - c_1)\frac{a_1 - p_1}{b_1}. \tag{A.4}$$

The first-order condition is $(a_1 + c_1 - 2p_1)/b_1 = 0$; the second derivative is $-2/b_1 < 0$. Hence

$$p_1^M = \frac{a_1 + c_1}{2}, \qquad q_1^M = Q^M = \frac{a_1 - c_1}{2b_1}. \tag{A.5}$$

Replacing c_1 by $c_1 - \delta$ gives

$$\Delta p_1^M = -\frac{\delta}{2}, \qquad \Delta q_1^M = \Delta Q^M = \frac{\delta}{2b_1}. \tag{A.6}$$

Half the saving lowers price; the other half raises the margin $p_1 - c_1$. Every additional NWC sale increases total purchases.

A.3 Price competition and business stealing

Prices. With two stores, prices are chosen simultaneously, each taking the other's price as given: differentiated Bertrand competition. Define $H = b_1 b_2 - d^2 > 0$. Solving (A.3) for quantities gives

$$q_1 = \frac{b_2(a_1 - p_1) - d(a_2 - p_2)}{H}, \qquad q_2 = \frac{b_1(a_2 - p_2) - d(a_1 - p_1)}{H}. \tag{A.7}$$

Thus $\partial q_1/\partial p_1 = -b_2/H$ and $\partial q_2/\partial p_2 = -b_1/H$. Differentiating $(p_i - c_i)q_i$ yields

$$q_i + (p_i - c_i)\frac{\partial q_i}{\partial p_i} = 0 \quad \Longrightarrow \quad q_1^D = \frac{b_2}{H}(p_1^D - c_1), \qquad q_2^D = \frac{b_1}{H}(p_2^D - c_2). \tag{A.8}$$

The second derivatives are $-2b_2/H$ and $-2b_1/H$, both negative. Substitute the demand equations into these conditions and collect the price terms:

$$\begin{aligned} p_1 &= \frac{a_1 + c_1}{2} - \frac{d}{2b_2}(a_2 - p_2), \\ p_2 &= \frac{a_2 + c_2}{2} - \frac{d}{2b_1}(a_1 - p_1). \end{aligned} \tag{A.9}$$

These best responses slope upward: each store cuts its price when the other does. Their slope product, $d^2/(4b_1 b_2) < 1$, ensures a unique interior price equilibrium.

Subtracting the pre-shock conditions from the post-shock conditions, with only NWC's cost changing, gives

$$\Delta p_1^D = -\frac{\delta}{2} + \frac{d}{2b_2}\Delta p_2^D, \qquad \Delta p_2^D = \frac{d}{2b_1}\Delta p_1^D. \tag{A.10}$$

Substitute the second equation into the first: $\left[1 - d^2/(4b_1 b_2)\right]\Delta p_1^D = -\delta/2$. Writing $K = 4b_1 b_2 - d^2 > 0$, the price changes are therefore

$$\Delta p_1^D = -\frac{2b_1 b_2}{K}\delta, \qquad \Delta p_2^D = -\frac{b_2 d}{K}\delta. \tag{A.11}$$

Both prices fall, with a larger reduction at NWC since $2b_1 > d$. NWC's pass-through rate, the price reduction per dollar of cost saving, satisfies

$$\tau_M = \frac{1}{2}, \qquad \tau_D = \frac{2b_1 b_2}{K}, \qquad \tau_D - \tau_M = \frac{d^2}{2K} > 0. \tag{A.12}$$

Competition adds a further price cut to the monopoly response: the rival responds to NWC's lower price, and NWC responds in turn.

Store sales and business stealing. Taking changes in (A.8), then using (A.11), gives

$$\begin{aligned} \Delta q_1^D &= \frac{b_2}{H}(\Delta p_1^D + \delta) = \frac{b_2(2b_1 b_2 - d^2)}{HK}\delta > 0, \\ \Delta q_2^D &= \frac{b_1}{H}\Delta p_2^D = -\frac{b_1 b_2 d}{HK}\delta < 0. \end{aligned} \tag{A.13}$$

The rival loses sales despite cutting its price. Define its lost sales as B, the net business-stealing component of NWC's gain:

$$B \equiv -\Delta q_2^D = \frac{b_1 b_2 d}{HK}\delta, \qquad \underbrace{\Delta q_1^D}_{\text{NWC's gain}} = \underbrace{\Delta Q^D}_{\text{additional purchases}} + \underbrace{B}_{\text{displaced rival sales}}. \tag{A.14}$$

This is substitution toward NWC with unchanged rival costs and freight access. Under monopoly, $B = 0$.

Community purchases. Adding the two quantity changes yields

$$\Delta Q^D = \frac{b_2(2b_1 b_2 - d^2 - b_1 d)}{HK}\delta > 0, \tag{A.15}$$

because $2b_1 b_2 - d^2 - b_1 d = H + b_1(b_2 - d) > 0$. Rival losses therefore offset part, but not all, of NWC's gain. Subtracting the monopoly response makes the competing forces explicit:

$$\begin{aligned} \Delta Q^D - \Delta Q^M &= \underbrace{\frac{d^2(3b_1 b_2 - d^2)}{2b_1 HK}\delta}_{\Delta q_1^D - \Delta q_1^M > 0} - \underbrace{\frac{b_1 b_2 d}{HK}\delta}_{B>0} \\ &= \frac{d(3b_1 b_2 d - 2b_1^2 b_2 - d^3)}{2b_1 HK}\delta. \end{aligned} \tag{A.16}$$

NWC expands more under competition, but some of that expansion displaces its rival. These opposing terms leave the ranking of *absolute* changes in community purchases ambiguous. If $b_1 = b_2 = b$, the expression simplifies to

$$\Delta Q^D = \frac{b\delta}{(b+d)(2b-d)} < \frac{\delta}{2b} = \Delta Q^M, \tag{A.17}$$

since $(b+d)(2b-d) = 2b^2 + d(b-d) > 2b^2$. Thus equal demand slopes give a smaller increase in total purchases under competition, alongside a larger fall in NWC's price. Unequal slopes can reverse the quantity

ranking: for $b_1 = 1$, $b_2 = 2$, and $d = 0.9$, $\Delta Q^D \simeq 0.535\delta > 0.5\delta = \Delta Q^M$.

Which sales receive the cost reduction? Let $s_1^0 = q_1^0/Q^0$ be NWC's initial share. At the initial sales mix, the delivered-cost reduction per unit of community purchases is

$$-\frac{q_1^0 \Delta c_1 + q_2^0 \Delta c_2}{Q^0} = s_1^0 \delta. \tag{A.18}$$

All sales receive the reduction under monopoly ($s_1^0 = 1$); only NWC's share receives it under duopoly ($0 < s_1^0 < 1$). Together with business stealing, this explains how a larger NWC price cut can coexist with a smaller increase in total purchases. The corresponding accounting identity for proportional changes is

$$\frac{\Delta Q}{Q^0} = s_1^0 \frac{\Delta q_1}{q_1^0} + (1 - s_1^0) \frac{\Delta q_2}{q_2^0}. \tag{A.19}$$

These comparisons are in levels; proportional changes also depend on initial sales. Initial shares vary with demand intercepts and costs, whereas τ_D depends only on slopes. Baseline share therefore measures cost exposure here; it does not independently determine pass-through.

A.4 Cost reductions at both retailers

Suppose freight improvements also lower the rival's cost by $\eta \geq 0$. The second price-change condition becomes $\Delta p_2^D = -\eta/2 + [d/(2b_1)]\Delta p_1^D$. Substitution into the first condition now gives $\left[1 - d^2/(4b_1 b_2)\right]\Delta p_1^D = -\delta/2 - d\eta/(4b_2)$, hence

$$\Delta p_1^D = -\frac{2b_1 b_2 \delta + b_1 d \eta}{K}, \qquad \Delta p_2^D = -\frac{b_2 d \delta + 2b_1 b_2 \eta}{K}. \tag{A.20}$$

Use $\Delta q_1^D = (b_2/H)(\Delta p_1^D + \delta)$ and $\Delta q_2^D = (b_1/H)(\Delta p_2^D + \eta)$ to obtain

$$\begin{aligned} \Delta q_1^D &= \frac{b_2[(2b_1 b_2 - d^2)\delta - b_1 d \eta]}{HK}, \\ \Delta q_2^D &= \frac{b_1[(2b_1 b_2 - d^2)\eta - b_2 d \delta]}{HK}. \end{aligned} \tag{A.21}$$

For equal savings ($\eta = \delta$), both numerators are positive: $2b_1 b_2 - d^2 - b_j d = H + b_j(b_i - d) > 0$ for $i \neq j$. Shared cost savings can therefore generate growth at both retailers.

A.5 Consumer surplus and total welfare

Consumer surplus is $CS = V - p_1 q_1 - p_2 q_2$, evaluated at consumers' chosen purchases. Differentiating and using $\partial V/\partial q_i = p_i$ cancels the quantity terms:

$$dCS = \sum_i \left(\frac{\partial V}{\partial q_i} - p_i \right) dq_i - \sum_i q_i \, dp_i = -\sum_i q_i \, dp_i. \tag{A.22}$$

Along the baseline NWC-only cost reduction, the price responses give

$$\frac{dCS^M}{d\delta} = \frac{q_1^M}{2} > 0, \qquad \frac{dCS^D}{d\delta} = \frac{2b_1b_2q_1^D + b_2dq_2^D}{K} > 0. \tag{A.23}$$

Consumers gain in both settings. The gain depends on both stores' price changes and sales. Total shipments measure the expansion in community purchases after netting out business stealing, but neither NWC's price change nor total quantity alone ranks monetary consumer gains across communities. This motivates examining NWC prices, retailer shipments, and total shipments together.

For total welfare, first consider a pure resource-cost reduction: c_i are the actual costs of supplying food and δ reduces those costs. Consumer surplus plus operating profits is then $W = CS + \pi_1 + \pi_2 = V - c_1q_1 - c_2q_2$. Since $dc_1/d\delta = -1$, $dc_2/d\delta = 0$, and $\partial V/\partial q_i = p_i$, differentiation gives

$$\frac{dW^D}{d\delta} = q_1^D + (p_1^D - c_1)\frac{dq_1^D}{d\delta} + (p_2^D - c_2)\frac{dq_2^D}{d\delta}. \tag{A.24}$$

The terms are the saving on NWC's current sales and the net benefits of the changes in each store's sales. From (A.8), margins are $p_1^D - c_1 = Hq_1^D/b_2$ and $p_2^D - c_2 = Hq_2^D/b_1$. Substituting these and the quantity responses gives

$$\begin{aligned}\frac{dW^D}{d\delta} &= q_1^D + \frac{2b_1b_2 - d^2}{K}q_1^D - \frac{b_2d}{K}q_2^D \\ &= \frac{2(3b_1b_2 - d^2)q_1^D - b_2dq_2^D}{K}.\end{aligned} \tag{A.25}$$

This is positive when $q_1^D/q_2^D > b_2d/[2(3b_1b_2 - d^2)]$. In particular, NWC initially selling at least as much as its rival is sufficient; this condition continues to hold as NWC expands and its rival contracts. Under monopoly, substituting $p_1^M - c_1 = b_1q_1^M$ and $dq_1^M/d\delta = 1/(2b_1)$ gives $dW^M/d\delta = 3q_1^M/2 > 0$.

Internalizing a freight markup changes NWC's pricing incentive without itself saving resources. A full welfare calculation must include upstream profits, so that freight payments are treated as transfers, alongside real delivery costs, subsidy payments, and investment costs. The consumer-surplus gain still follows from lower prices.

B Additional Empirical Results

B.1 RNFB conditioning and the common-sample comparison

Table B.1 adds log RNFB price to the retailer-level shipment regressions. Table B.2 uses the same 5,208 observations without that price control. This separates the change associated with conditioning on price from the change associated with limiting the sample to observations with RNFB data. In both tables, NWC expands on expansion routes while the local-retailer estimate is negative and imprecise. The small same-sample changes do not establish that price changes cause the measured retailer reallocation.

Table B.1: Retailer-level decomposition controlling for RNFB price

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
Panel A. All NSA routes				
Post × NSA	2.378***	2.634***	0.011	−0.256
	(0.785)	(0.890)	(0.775)	(0.790)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.886	0.834
Panel B. Original vs. expansion routes				
Post × original NSA	3.520***	1.659	2.970***	1.860
	(1.209)	(1.346)	(1.054)	(1.137)
Post × expansion	2.103**	2.869***	−0.701	−0.766
	(0.827)	(0.863)	(0.701)	(0.749)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.889	0.835
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Log RNFB price control	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: All outcomes are shipment kilograms per resident per reporting period. Column (4) includes the local-retailer shipments in column (3), so these columns must not be added together. Column (1) is total shipments, column (2) is NWC, and column (4) is all non-NWC shipments. Panel A pools routes; Panel B includes original and expansion interactions jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. This specification additionally controls for log RNFB price. Compare with Table B.2, which uses the same sample without that control. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table B.2: Retailer-level decomposition on the same sample without RNFB price

	Total shipments (1)	NWC shipments (2)	Other local retailers (3)	Other retailers + direct orders (4)
Panel A. All NSA routes				
Post × NSA	2.501***	2.870***	−0.021	−0.369
	(0.790)	(0.837)	(0.766)	(0.788)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.886	0.834
Panel B. Original vs. expansion routes				
Post × original NSA	3.673***	1.995	2.887***	1.678
	(1.224)	(1.322)	(1.032)	(1.126)
Post × expansion	2.204***	3.092***	−0.756	−0.887
	(0.827)	(0.815)	(0.692)	(0.742)
Observations	5,208	5,208	5,208	5,208
Adjusted R^2	0.641	0.890	0.888	0.835
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Subsidy/kg control	Yes	Yes	Yes	Yes
Log RNFB price control	No	No	No	No
RNFB-observed sample	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: All outcomes are shipment kilograms per resident per reporting period. Column (4) includes the local-retailer shipments in column (3), so these columns must not be added together. Column (1) is total shipments, column (2) is NWC, and column (4) is all non-NWC shipments. Panel A pools routes; Panel B includes original and expansion interactions jointly. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. The sample is the same RNFB-observed sample as Table B.1, but log RNFB price is not included as a regressor. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.2 Retailer-specific proportional changes

Table B.3 reports pooled log-shipment estimates of 0.125 for NWC and 0.168 for other retailers, significant at the 10% and 5% levels. The columns contain 9,240 and 8,944 observations, respectively. These are proportional comparisons on positive-shipment samples, not an additive decomposition of total shipments.

Table B.3: Effects on retailer-specific log shipment quantities

	Log NWC shipments (1)	Log other-retailer shipments (2)
Post × NSA	0.125*	0.168**
	(0.065)	(0.084)
Observations	9,240	8,944
Adjusted R^2	0.934	0.863
Community fixed effects	Yes	Yes
Period fixed effects	Yes	Yes
Subsidy/kg control	Yes	Yes
Baseline rate × period	Yes	Yes

Notes: The dependent variables are log shipment weight associated with NWC and other retailers. Treatment pools all NSA route communities. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. The per-kilogram subsidy is controlled for. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. Logarithms require positive retailer-specific shipment weights, and the two columns have different estimation samples. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.3 Original versus expansion route effects

Table B.4 reports coefficient-equality tests. The price difference has a p-value of 0.279. For the quantity estimates of 0.124 and 0.083, the p-value is 0.196. Neither null is rejected at conventional levels. The quantity test does not apply to the different estimates reproduced in Table 3.

Table B.4: Tests of differences between original and expansion route effects

			Difference		
	Original	Expansion	Original − expansion	F-statistic	p-value
Log RNFB price	−0.053	−0.035	−0.017	1.19	0.279
Log shipment quantity	0.124	0.083	0.041	1.69	0.196

Notes: The null is equality of the original- and expansion-route coefficients. The displayed coefficients and differences retain the source rounding. The quantity test concerns the 0.124 and 0.083 estimates in Table 6, not the 0.211 and 0.062 estimates in Table 3. Price regressions control for the RNFB subsidy; quantity regressions control for the per-kilogram subsidy. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Inference uses community-clustered standard errors.

B.4 Shipment-weight cutoffs

Table B.5 reports the cutoff checks discussed in Section 8. Each positive cutoff applies jointly to the community's mean shipment weight per resident and its contemporaneous value.

Table B.5: Sensitivity to shipment-weight cutoffs

	No cutoff	> 10	> 15	> 20
	(1)	(2)	(3)	(4)
Panel A. Log RNFB price: all NSA routes				
Post × NSA	−0.037**	−0.037**	−0.039**	−0.035**
	(0.014)	(0.014)	(0.015)	(0.015)
Observations	5,348	5,314	5,219	4,672
Adjusted R^2	0.934	0.934	0.936	0.936
Panel B. Log RNFB price: original vs. expansion routes				
Post × original NSA	−0.053***	−0.052***	−0.053***	−0.044**
	(0.019)	(0.019)	(0.020)	(0.021)
Post × expansion	−0.033**	−0.033**	−0.035**	−0.033**
	(0.015)	(0.015)	(0.015)	(0.015)
Observations	5,348	5,314	5,219	4,672
Adjusted R^2	0.934	0.934	0.936	0.936
Panel C. Log shipment quantity: all NSA routes				
Post × NSA	0.125***	0.108***	0.095***	0.066***
	(0.046)	(0.027)	(0.022)	(0.019)
Observations	12,975	12,495	11,727	9,712
Adjusted R^2	0.906	0.960	0.965	0.970
Panel D. Log shipment quantity: original vs. expansion routes				
Post × original NSA	0.223***	0.155***	0.124***	0.093***
	(0.074)	(0.039)	(0.033)	(0.027)
Post × expansion	0.076*	0.087***	0.083***	0.054**
	(0.040)	(0.026)	(0.023)	(0.021)
Observations	12,975	12,495	11,727	9,712
Adjusted R^2	0.906	0.960	0.965	0.970
Community fixed effects	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: Column (1) imposes no shipment-weight cutoff. Columns (2)–(4) require both mean and contemporaneous shipment kilograms per resident to exceed 10, 15, or 20, respectively. Column (3) is the benchmark for this exercise. All columns exclude the phase-in period. Price regressions control for the RNFB subsidy and quantity regressions for the per-kilogram subsidy. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.5 Alternative shipment coverage

Table B.6 compares the benchmark shipment measure with the broader all-NNC-shipment measure, using the corresponding subsidy control in each column.

Table B.6: Robustness to an alternative shipment quantity measure

	Benchmark (1)	Alternative: all NNC shipments (2)
Panel A. All NSA routes		
Post × NSA	0.095***	0.130***
	(0.022)	(0.026)
Observations	11,727	11,727
Adjusted R^2	0.965	0.962
Panel B. Original vs. expansion routes		
Post × original NSA	0.124***	0.183***
	(0.033)	(0.043)
Post × expansion	0.083***	0.107***
	(0.023)	(0.027)
Observations	11,727	11,727
Adjusted R^2	0.965	0.962
Community fixed effects	Yes	Yes
Period fixed effects	Yes	Yes
Baseline rate × period	Yes	Yes
Quantity subsidy control	Benchmark	Alternative

Notes: Column (1) uses log benchmark shipment weight and its per-kilogram subsidy control. Column (2) uses the log of the broader measure covering all NNC shipments and the corresponding alternative subsidy control. Panel A pools all routes; Panel B includes original and expansion routes jointly. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. All specifications include community and period fixed effects and baseline-subsidy-rate-by-period controls. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.6 Community Specific seasonality

Table B.7 replaces ordinary community fixed effects with community-by-quarter-of-year effects while retaining calendar-period fixed effects. It evaluates sensitivity to recurring local seasonal patterns.

Table B.7: Robustness to community-specific seasonality

	All NSA routes		Original vs. expansion	
	Benchmark	Seasonality	Benchmark	Seasonality
	(1)	(2)	(3)	(4)
Panel A. Log RNFB price				
Post × NSA	−0.039**	−0.038**		
	(0.015)	(0.015)		
Post × original NSA			−0.053***	−0.052**
			(0.020)	(0.020)
Post × expansion			−0.035**	−0.035**
			(0.015)	(0.015)
Observations	5,219	5,217	5,219	5,217
Adjusted R^2	0.936	0.935	0.936	0.935
Panel B. Log shipment quantity				
Post × NSA	0.095***	0.090***		
	(0.022)	(0.023)		
Post × original NSA			0.124***	0.118***
			(0.033)	(0.033)
Post × expansion			0.083***	0.079***
			(0.023)	(0.023)
Observations	11,727	11,727	11,727	11,727
Adjusted R^2	0.965	0.966	0.965	0.966
Community effects absorbed	Yes	Yes	Yes	Yes
Period fixed effects	Yes	Yes	Yes	Yes
Community × quarter-of-year	No	Yes	No	Yes
Baseline rate × period	Yes	Yes	Yes	Yes

Notes: Columns (1) and (3) use standard community fixed effects. Columns (2) and (4) replace them with community-by-quarter-of-year fixed effects, allowing recurring community-specific Q1–Q4 patterns; these also absorb community main effects. Calendar-period fixed effects and baseline-rate-by-period interactions remain in every column. Price regressions control for the RNFB subsidy and quantity regressions for the per-kilogram subsidy. The reported sample requires both mean and contemporaneous shipment weight per resident to exceed 15 kilograms and excludes the phase-in period. Standard errors clustered by community are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.7 Formal pre-treatment trend tests

Table B.8 compares average event-study coefficients across adjacent pre-treatment windows of 3, 6, and 12 months. None of the tests rejects equality at the 5% level, while three price tests reject at the 10% level.

Table B.8: Formal tests of pre-treatment trends

	12-month blocks	6-month blocks	3-month blocks
Panel A. RNFB price			
All NSA routes	0.206	0.064	0.931
Original NSA routes	0.656	0.071	0.222
Expansion NSA routes	0.051	0.257	0.503
Panel B. Shipment kilograms per resident			
All NSA routes	0.711	0.145	0.217
Original NSA routes	0.838	0.310	0.174
Expansion NSA routes	0.794	0.188	0.337

Notes: Entries are p-values from tests comparing average event-study coefficients across adjacent pre-treatment windows of 12, 6, and 3 months. Price estimates use quarterly RNFB costs (March 2017 omitted); shipment estimates use monthly kilograms per resident (June 2017 omitted). Regressions include community and period fixed effects, subsidy controls, and baseline-rate-by-period interactions. Standard errors are clustered by community. Price observations: 3,618; quantity observations: 12,916.

B.8 Continuous baseline NWC shipment share

Table B.9 replaces the high-share indicator with NWC's continuous baseline shipment share. The original-route price interaction remains positive and statistically significant, while the expansion-route quantity interaction is positive and significant at the 10% level.

Table B.9: Robustness: continuous baseline NWC shipment share

	All NSA routes (1)	Original vs. expansion (2)
Panel A. Log RNFB price		
Post × NSA	−0.126	
	(0.087)	
Post × NSA × share	0.150	
	(0.107)	
Post × original NSA routes		−0.285***
		(0.087)
Post × original × share		0.301***
		(0.106)
Post × expansion routes		−0.127
		(0.100)
Post × expansion × share		0.159
		(0.125)
Observations	5,219	5,219
Adjusted R^2	0.939	0.939
Panel B. Log total shipment weight		
Post × NSA	0.039	
	(0.058)	
Post × NSA × share	0.078	
	(0.084)	
Post × original NSA routes		0.103
		(0.069)
Post × original × share		0.044
		(0.092)
Post × expansion routes		−0.090
		(0.100)
Post × expansion × share		0.226*
		(0.135)
Observations	11,727	11,727
Adjusted R^2	0.965	0.965
Community fixed effects	Yes	Yes
Period fixed effects	Yes	Yes
Outcome-specific subsidy	Yes	Yes
Baseline rate × period	Yes	Yes

Notes: Continuous baseline NWC shipment share ranges from zero to one and replaces the high-share indicator in Table 4. Treatment coefficients are evaluated at zero share; interaction coefficients measure how treatment effects vary with share. Zero-share communities are included. Regressions include community and period fixed effects, outcome-specific subsidies, and baseline-rate-by-period interactions. The 15 kg/person sample restrictions apply, and the phase-in period is excluded. Standard errors are clustered by community. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

B.9 Excluding zero-share communities

Table B.10 repeats the high/low-share analysis among communities with positive baseline NWC shipment shares. The original-route price interaction remains positive and statistically significant, and high-share shipment effects are positive and significant at the 10% level across the pooled and route-specific specifications.

Table B.10: Robustness: high and low baseline NWC shipment shares among positive-share communities

	Regression coefficients		High-share joint effects	
	All NSA (1)	Original/expansion (2)	All NSA (3)	Original/expansion (4)
Panel A. Log RNFB price				
Post × NSA	−0.066**		−0.014	
	(0.027)		(0.021)	
Post × NSA × high share	0.052			
	(0.035)			
Post × original NSA		−0.109***		−0.030
		(0.018)		(0.024)
Post × original × high share		0.079***		
		(0.025)		
Post × expansion		−0.063**		−0.008
		(0.028)		(0.021)
Post × expansion × high share		0.055		
		(0.036)		
Observations	4,861	4,861		
Adjusted R^2	0.933	0.934		
Panel B. Log total shipment weight				
Post × NSA	0.028		0.097*	
	(0.033)		(0.055)	
Post × NSA × high share	0.068			
	(0.068)			
Post × original NSA		0.125***		0.104*
		(0.027)		(0.061)
Post × original × high share		−0.021		
		(0.052)		
Post × expansion		0.022		0.096*
		(0.034)		(0.055)
Post × expansion × high share		0.074		
		(0.069)		
Observations	9,199	9,199		
Adjusted R^2	0.959	0.959		
Community fixed effects	Yes	Yes		
Period fixed effects	Yes	Yes		
Outcome-specific subsidy	Yes	Yes		
Baseline rate × period	Yes	Yes		

Notes: The sample excludes zero and missing baseline NWC shipment shares. High share exceeds 70%; low share is positive and at or below 70%. Columns (1)–(2) report low-share effects and high-minus-low interactions; columns (3)–(4) report high-share joint effects. Regressions include community and period fixed effects, outcome-specific subsidy controls, and baseline-rate-by-period interactions. Both samples apply the 15 kg/person restrictions and exclude the phase-in period. Community-clustered standard errors are in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.